

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

01

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

For a person m miles from a flash of lightning, the length of the time interval from the moment the person sees the lightning to the moment the person hears the thunder is k seconds. The ratio of m to k can be estimated to be 1 to 5. According to this estimate, the person is how many miles from a flash of lightning if the time interval is 25 seconds?

- A. 10
- B. 9
- C. 6
- D. 5

Q2

PROBLEM-SOLVING AND DATA ANALYSIS

In a study, the data from a random sample of a population had a mean of 37, with an associated margin of error of 3. Which of the following is the most appropriate conclusion that can be made about the population mean?

- A. It is less than 37.
- B. It is greater than 37.
- C. It is between 34 and 40.
- D. It is less than 34 or greater than 40.

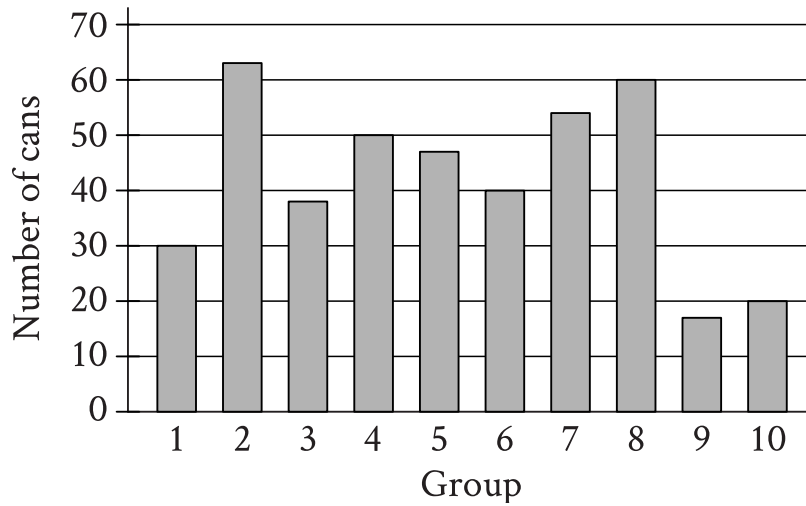
There were approximately 113,000 occupational therapy jobs in the United States in 2012. The Bureau of Labor Statistics has projected that this number will increase by 29% from 2012 to 2022. Of the following, which is closest to the number of occupational therapy jobs the bureau has projected for the United States in 2022?

- A. 115,900
- B. 116,300
- C. 142,000
- D. 145,800

Voice type	Number of singers
Countertenor	4
Tenor	6
Baritone	10
Bass	5

A total of 25 men registered for singing lessons. The frequency table shows how many of these singers have certain voice types. If one of these singers is selected at random, what is the probability he is a baritone?

- A. 0.10
- B. 0.40
- C. 0.60
- D. 0.67



- The data for the 10 categories are as follows:

- Group 1: 30
- Group 2: 63
- Group 3: 38
- Group 4: 50
- Group 5: 47
- Group 6: 40
- Group 7: 54
- Group 8: 60
- Group 9: 17
- Group 10: 20

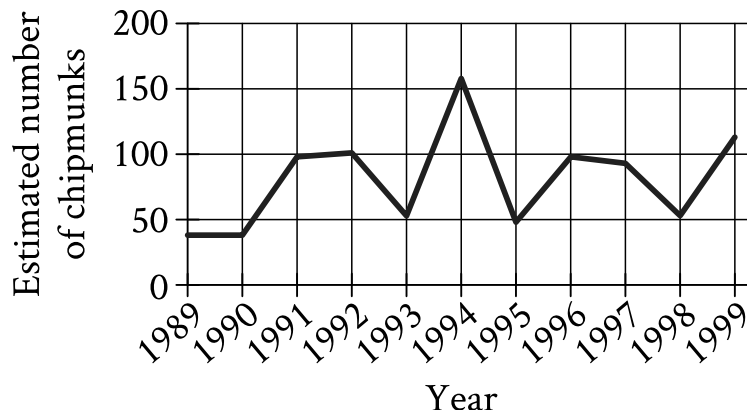
The bar graph shows the distribution of 419 cans collected by 10 different groups for a food drive. How many cans were collected by group 6?

STUDENT-PRODUCED RESPONSE

☐ 3	☐ 3	☐ 3	☐ 3
☐ 4	☐ 4	☐ 4	☐ 4
☐ 5	☐ 5	☐ 5	☐ 5
☐ 6	☐ 6	☐ 6	☐ 6
☐ 7	☐ 7	☐ 7	☐ 7
☐ 8	☐ 8	☐ 8	☐ 8
☐ 9	☐ 9	☐ 9	☐ 9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The line graph shows the estimated number of chipmunks in a state park on April 1 of each year from 1989 to 1999.



- The line graph:
 - Begins at 1989, 38 chipmunks
 - Remains level to 1990, 38 chipmunks
 - Rises sharply to 1991, 98 chipmunks
 - Rises gradually to 1992, 101 chipmunks
 - Falls sharply to 1993, 53 chipmunks
 - Rises sharply to 1994, 158 chipmunks
 - Falls sharply to 1995, 48 chipmunks
 - Rises sharply to 1996, 98 chipmunks
 - Falls gradually to 1997, 93 chipmunks
 - Falls sharply to 1998, 53 chipmunks
 - Rises sharply to 1999, 113 chipmunks

Based on the line graph, in which year was the estimated number of chipmunks in the state park the greatest?

- A. 1989
- B. 1994
- C. 1995

D. 1998

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

The members of a city council wanted to assess the opinions of all city residents about converting an open field into a dog park. The council surveyed a sample of 500 city residents who own dogs. The survey showed that the majority of those sampled were in favor of the dog park. Which of the following is true about the city council's survey?

- A. It shows that the majority of city residents are in favor of the dog park.
- B. The survey sample should have included more residents who are dog owners.
- C. The survey sample should have consisted entirely of residents who do not own dogs.
- D. The survey sample is biased because it is not representative of all city residents.

Q8

PROBLEM-SOLVING AND DATA ANALYSIS

How many yards are equivalent to 612 inches? 1 yard = 36 inches

- A. 0.059
- B. 17
- C. 576
- D. 22,032

Q9

PROBLEM-SOLVING AND DATA ANALYSIS

There are 55 students in Spanish club. A sample of the Spanish club students was selected at random and asked whether they intend to enroll in a new study program. Of those surveyed, 20% responded that they intend to enroll in the study program. Based on this survey, which of the following is the best estimate of the total number of Spanish club students who intend to enroll in the study program?

- A. 11
- B. 20
- C. 44
- D. 55

Q10

PROBLEM-SOLVING AND DATA ANALYSIS

What is 10% of 470?

- A. 37
- B. 47
- C. 423
- D. 460